Opposite Encoding Preferences for Quantum-Record Consensus and Pooled Inference

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Abstract

Can an encoding that best supports separated observers' simultaneous correctness be worse when their quantum records are read together? The abstract possibility is already present in classical simultaneous discrimination. We resolve a constrained physical version: two pure product encodings pass through the same common random-rotation channel and deliver identical conditional one-record density operators. We classify every pure-product encoder with these prescribed marginals, up to fixed local unitary frames, into two classes. An attained norm bound solves all local binary measurements, including private randomness and shared classical seeds. Throughout an open signal/noise region, one class strictly maximizes all-correct minimax consensus while the other maximizes pooled binary inference. At an exact benchmark their consensus gap is 1/16; both scores exceed a shared random guess. The pooled optimum is attained by one-way communication and adaptive local measurements. A second realization uses equal-energy coherent records under common Gaussian displacement. An exact latent-amplitude relaxation, a weighted-path reduction and rational certificates prove a strict reversal for arbitrary local quantum measurements without a Fock cutoff. The result concerns operational encoding choices under correlated preparation noise, not a new theory of objectivity, an entanglement advantage or an experimental demonstration.

1 Question and relation to prior work

Environmental records can make selected system information available to many observers [1, 2]. Quantum Darwinism, spectrum broadcast structure and related frameworks distinguish several conditions involved in that process [3, 4]. Operational notions of consensus need not coincide [5, 6]; local parameter-estimation precision supplies another criterion [7]. Here the source variable is a specified binary label, rather than an observable whose selection must be derived.

We compare two tasks. In *separated consensus*, every observer must identify the actual label correctly, without communication. In *pooled inference*, one decision is made from all records. Consensus means correctness against the label, not mere agreement. Local simultaneous state discrimination (LSSD) already formalizes an all-correct task [8, 9]; we do not introduce it. Indeed, classical LSSD already permits opposite task rankings at identical local marginals. A compact exact classical example is given in Appendix A, to delimit the claim.

The question addressed here is more constrained: *which encodings optimize these two tasks through one common-noise channel, when the complete local density-operator pairs are held fixed?* We solve this for a two-qubit class. The result includes a complete encoder classification under explicit pure-input and fixed-frame assumptions, a global all-POVM optimum, an open two-parameter reversal region, and perturbation margins. A Gaussian realization shows that the optimized separation is not tied to a two-level calculation or a convenient finite cutoff.

Common-noise-resistant encoding is established physics. Parallel and antiparallel spin encodings already have different pooled information properties [10]; our two angle encodings are not a universal antiparallel spin flip. Common Gaussian displacement channels and homodyne comparisons also predate this work [11]. Neither correlation sensitivity nor the generic failure of local diagnostics to determine a joint task is claimed as new. The candidate contribution is the solved matched-channel quantum family and the unrestricted-decoder certificates under these constraints. A bounded primary-source comparison did not locate the same construction; this is not a proof of priority. Related unpublished work cited by Majenz et al. as "Friend or foe in quantum state discrimination" was not obtained in full, so that coverage limit remains explicit.

2 Tasks and resources

Let $x \in \{+1, -1\}$ label two known conditional states ρ_x on two record systems. Each observer returns one label on every trial. With local binary effects E, F for reporting $+1$, put

$$A(E, F) = \operatorname{Tr} \rho_+ (E \otimes F), \tag{1}$$

$$B(E, F) = \operatorname{Tr} \rho_- [(\mathbb{I} - E) \otimes (\mathbb{I} - F)], \qquad 0 \preceq E, F \preceq \mathbb{I}. \tag{2}$$

The private-product minimax confidence is $C_{\text{priv}} = \sup_{E,F} \min(A, B)$. The shared-seed value C_{sh} additionally permits a classical mixture of product decoders. Its seed is independent of the *entire preparation*, including both x and the hidden noise. No communication, entangled ancillary resource, source-dependent advice or postselection is allowed. All local quantum instruments followed by a binary report are included by their effects.

The equal-prior product average is $J(E, F) = (A + B)/2$. Always $C_{\text{priv}} \le C_{\text{sh}} \le \sup J$. If a local unitary exchanges the two source states, a fair common seed choosing a decoder and its label-swapped conjugate gives both branches success J. Thus $C_{\text{sh}} = \sup J$ in our symmetric families. Where the maximizing product decoder already balances A and B, all three values coincide.

Pooled equal-prior success is the Helstrom value

$$P(\rho) = \frac{1}{2} + \frac{1}{4} \|\rho_+ - \rho_-\|_1. \tag{3}$$

In the symmetric models a pooled minimax interpretation gives the same value. Pooled access is a different information resource from noncommunicating local decisions; it must not be silently treated as an objectivity certificate.

3 A complete matched-marginal qubit encoding problem

Write $|\psi(a)\rangle = \cos(a/2)|0\rangle + \sin(a/2)|1\rangle$, and let $R_y(a) = e^{-iaY/2}$. Fix $0 < \phi < \pi/4$. The same source-independent random-unitary channel acts on either input:

$$\mathcal{N}_\phi(\omega) = \frac{1}{2} \sum_{h=\pm 1} [R_y(h\phi) \otimes R_y(h\phi)] \omega [R_y(h\phi) \otimes R_y(h\phi)]^\dagger. \tag{4}$$

The latent sign h is unavailable to the decoders.

We allow arbitrary pure product input states for each label and arbitrary fixed local unitary output frames, independent of x, h. Require that in the reported local frames the conditional Bloch vectors at *both* sites be

$$\mathbf{r}_x = (xd, 0, d), \qquad d = \frac{\cos\phi}{\sqrt{2}}. \tag{5}$$

This fixes the complete local discrimination experiments, not just one entropy or accuracy score.

Theorem 1 (Encoder classification and opposite task optima)**.** *Every allowed encoder/frame pair in the specified class produces, in the canonical local frames, one of two conditional ensembles, up to reversal of the latent sign:*

$$\rho_x^\kappa = \frac{1}{2}\sum_{h=\pm1} |\psi(x\pi/4+h\phi)\rangle\langle\psi(x\pi/4+h\phi)| \otimes |\psi(x\pi/4+\kappa h\phi)\rangle\langle\psi(x\pi/4+\kappa h\phi)|, \quad \kappa=\pm1. \tag{6}$$

Define $t_+ = 1/2$, $t_- = \cos(2\phi)/2$, $k_+ = \cos(2\phi)/2$, $k_- = 1/2$. *Their exact task values are*

$$C_\kappa = C_{\text{priv}} = C_{\text{sh}} = \frac{1+2d+t_\kappa}{4}, \qquad P_\kappa = \frac{1+\sqrt{d^2+k_\kappa^2}}{2}. \tag{7}$$

Consequently $C_+ > C_- > 1/2$, *whereas* $P_- > P_+$. *Thus the optimal encoder class is different for the two tasks over the whole specified encoding class.*

3.1 Why the classification is complete

The marginal channel acts on Bloch vectors by $M = \text{diag}(c, 1, c)$, $c = \cos\phi$. For a pure input vector $\mathbf{n}$,

$$\|M\mathbf{n}\|^2 = c^2 + (1-c^2)n_y^2. \tag{8}$$

The prescribed outputs have length c. Unitary frames preserve length, so every allowed input has $n_y = 0$. Both conditional output vectors lie in the XZ plane and are noncollinear: the prescribed pair spans that plane. A fixed unitary frame mapping this pair to (5) must preserve the plane and send its normal y to ϵy, $\epsilon = \pm1$. Conjugation therefore sends $R_y(h\phi)$ to $R_y(\epsilon h\phi)$. Since M equals multiplication by c on that plane, the canonical pre-noise pure vectors are forced to be $(x/\sqrt{2}, 0, 1/\sqrt{2})$. At two sites only the signs ϵ_1, ϵ_2 remain; globally reversing h leaves only their relative sign κ. This proves (6) without assuming the two encodings exhaust the possibilities in advance.

Concrete representatives prepare angles $(x\pi/4, x\pi/4)$ or $(x\pi/4, -x\pi/4)$ before (4); in the second case a fixed Z frame on the second record gives (6). These representatives have equal excitation energies before and after noise, and total negative-log squared local-label overlap $2\ln 2$. That overlap is fixed throughout the classified pure-input class. General unitary frame representatives need not conserve physical Z energy. The energy statement concerns the explicit witnesses, or an energy-matched subclass containing them; it is not a claim that arbitrary frame changes are thermodynamically free. Mixed or entangled inputs, nonunitary output processing, frame choices depending on x, h, and flagged encoder mixtures are outside the classification.

3.2 Global optimization over local measurements

For any binary two-qubit ensemble set $\Omega = (\rho_+ + \rho_-)/2$, $D = (\rho_+ - \rho_-)/2$, and define

$$a_i = \text{Tr}\, D(\sigma_i \otimes \mathbb{I}), \quad b_i = \text{Tr}\, D(\mathbb{I} \otimes \sigma_i), \quad T_{ij} = \text{Tr}\, \Omega(\sigma_i \otimes \sigma_j).$$

The average objective is separately affine in E, F; compactness gives a maximizer with each effect extreme. Qubit extreme effects are $0, \mathbb{I}$ and rank-one projectors. Any constant-effect case has $J \le 1/2$. For projectors with unit Bloch axes $\mathbf{n}, \mathbf{m}$,

$$J = \frac{1+\mathbf{a}\cdot\mathbf{n}+\mathbf{b}\cdot\mathbf{m}+\mathbf{n}^T T\mathbf{m}}{4} \le \frac{1+\|\mathbf{a}\|+\|\mathbf{b}\|+\|T\|_{\text{op}}}{4}. \tag{9}$$

This bound is not asserted tight for arbitrary ensembles.

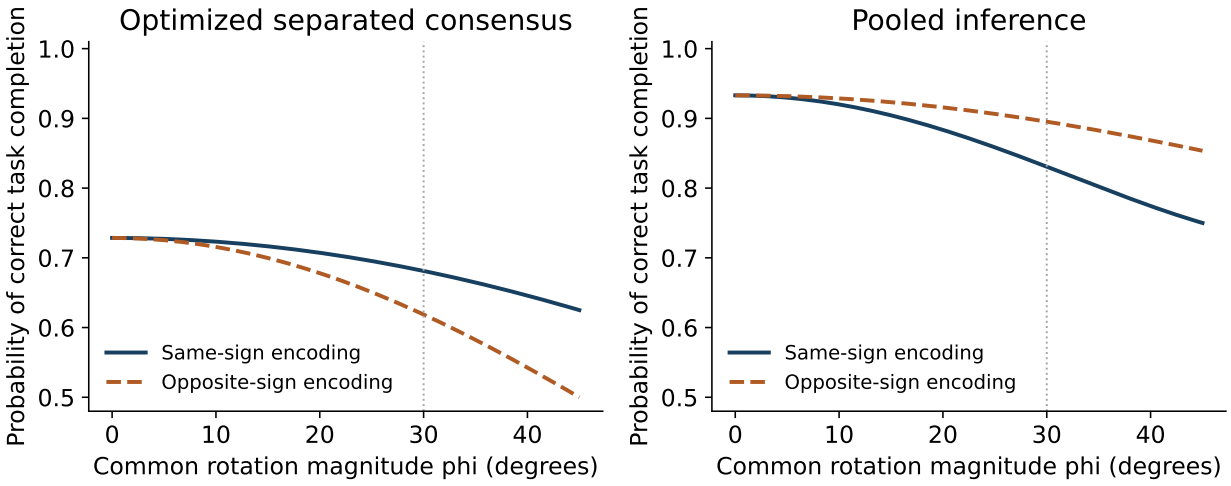


Figure 1: Exact optimized values for the complete classified qubit encoding problem. The noise magnitude ϕ varies while the two encodings have identical local density operators at each point. The vertical marker is the radical benchmark $\phi = \pi/6$. All curves evaluate analytical formulas; no device measurements or optimizer estimates enter them.

For (6), direct expansion gives

$$\rho_x^\kappa = \frac{1}{4}\{\mathbb{I}\otimes\mathbb{I} + d(Z\otimes\mathbb{I} + \mathbb{I}\otimes Z) + xd(X\otimes\mathbb{I} + \mathbb{I}\otimes X) + t_\kappa(X\otimes X + Z\otimes Z) + xk_\kappa(X\otimes Z + Z\otimes X)\}. \tag{10}$$

Hence $\mathbf{a} = \mathbf{b} = d\mathbf{e}_x$ and $T = t_\kappa \operatorname{diag}(1,0,1)$, with $t_\kappa \geq 0$. The X projectors attain every term of (9) simultaneously and give equal branch successes. This proves the consensus formula in (7) over all complex, biased and nonprojective local effects and their source-independent shared mixtures.

The difference of the two conditional states is

$$\Delta_\kappa = \frac{1}{2}[d(X\otimes\mathbb{I} + \mathbb{I}\otimes X) + k_\kappa(X\otimes Z + Z\otimes X)]. \tag{11}$$

Its eigenvalues are $\pm\sqrt{d^2 + k_\kappa^2}, 0, 0$, giving the pooled formula. Finally

$$C_+ - C_- = \frac{\sin^2\phi}{4} > 0, \qquad k_- > k_+ \geq 0 \implies P_- > P_+. \tag{12}$$

Both sources are separable mixtures of product pure states, but are not conditionally product. Their correlations are created by the hidden common noise.

At $\phi = \pi/6$ the values are

Encoding class	Optimized consensus	Pooled inference
Same sign	$(3+\sqrt{6})/8$	$1/2+\sqrt{7}/8$
Opposite sign	$(5+2\sqrt{6})/16$	$1/2+\sqrt{10}/8$

Numerically these are $(0.681186, 0.830719)$ and $(0.618686, 0.895285)$, respectively. The consensus gap is exactly $1/16$. Replacing either correlated conditional state by the product of its marginals instead gives consensus $(11+4\sqrt{6})/32 \simeq 0.649936$, halfway between the two actual values. Marginal quality therefore does not even determine which side of that product prediction the optimized consensus occupies.

Corollary 2 (An open region of signal and noise angles)**.** *The reversal extends throughout* $\pi/4 \le \theta < \pi/2$, $0 < \phi < \pi/4$, *with prescribed local Bloch pairs* $(x\sin\theta\cos\phi, 0, \cos\theta\cos\phi)$ *and pure inputs* $|\psi(x\theta)\rangle$. *The same two-class encoding classification holds. Put* $d_\theta = \sin\theta\cos\phi$ *and*

$$t_\kappa^x = \sin^2\theta\cos^2\phi + \kappa\cos^2\theta\sin^2\phi, \qquad h_+ = \cos 2\phi, \quad h_- = 1.$$

The exact optimized scores are

$$C_\kappa = \frac{1 + 2d_\theta + t_\kappa^x}{4}, \qquad P_\kappa = \frac{1 + \sin\theta\sqrt{\cos^2\phi + \cos^2\theta\, h_\kappa^2}}{2}.$$

In particular, $C_+ - C_- = \cos^2\theta\sin^2\phi/2 > 0$, $C_- > 1/2$, *and* $P_- > P_+$.

Proof. The marginal channel still forces the two noncollinear pure inputs into the XZ plane, giving the same frame classification. The average correlation tensor is $\mathrm{diag}(t_\kappa^x, 0, t_\kappa^z)$, where $t_\kappa^z = \cos^2\theta\cos^2\phi + \kappa\sin^2\theta\sin^2\phi$. In the stated region $t_\kappa^x \ge |t_\kappa^z|$, so the norm bound is again attained by X projectors. Indeed the difference of these two diagonal entries is $-\cos 2\theta\cos 2\phi \ge 0$ for $\kappa = +$ and $-\cos 2\theta \ge 0$ for $\kappa = -$, while their sums are respectively 1 and $\cos 2\phi > 0$. The conditional-state difference has the form (11) with $d = d_\theta$ and $k = \sin\theta\cos\theta h_\kappa$, giving the pooled formula. Finally $\partial_\theta C_- = \cos\theta(\cos\phi + \sin\theta)/2 > 0$, so its value exceeds the already informative $\theta = \pi/4$ baseline. Both displayed witnesses retain identical marginals and equal energies at each fixed parameter pair; no general control-work claim is added. □

3.3 Pooled inference needs only one-way communication

No entangling measurement is required for the pooled value. Alice measures X and sends her outcome $a = \pm 1$ to Bob. His unnormalized conditional difference is

$$D_a = \mathrm{Tr}_A[(((\mathbb{I} + aX)/2) \otimes \mathbb{I})\Delta_\kappa] = \frac{1}{2}[ad\mathbb{I} + dX + ak_\kappa Z]. \tag{13}$$

Its trace norm is $r = \sqrt{d^2 + k_\kappa^2}$. Bob measures along $(dX + ak_\kappa Z)/r$ and uses that outcome as the pooled guess. The sum of the two conditional trace norms is $2r = \|\Delta_\kappa\|_1$, attaining (3), without discarding any outcome. Alice's original measurement outcome is not required to equal that final guess. Broadcasting Bob's guess would change the separated task by allowing communication. By contrast, measuring X at both sites and merely pooling those fixed classical outcomes gives success $(1 + d)/2$ for either class; it misses the ordering carried by adaptive quantum readout.

3.4 Robustness

If each full conditional state is changed by trace distance at most η, the success of any fixed admissible protocol changes by at most η. Taking branch minima or optimizing over a fixed strategy class preserves that bound, including shared mixtures. Therefore both orderings at the benchmark persist for $\eta < 1/32$: the consensus gap is $1/16$, and the pooled gap $(\sqrt{10} - \sqrt{7})/8$ is larger. This is an open neighborhood result, not a claim that perturbed sources retain exactly identical marginals.

4 A Gaussian realization with unrestricted-decoder bounds

The same operational question has an infinite-dimensional realization under the established common-displacement channel [11]. For two modes, real $\alpha > 0$, complex Gaussian z of variance n_c,

and signs $\epsilon_r = \pm 1$, put

$$\rho_x^\epsilon = \int \frac{d^2 z}{\pi n_c} e^{-|z|^2/n_c} \bigotimes_{r=1}^{2} |x\epsilon_r \alpha + z\rangle\langle x\epsilon_r \alpha + z|. \tag{14}$$

Aligned signs are $(+,+)$; balanced signs are $(+,-)$. Known local parity aligns the marginal label conventions. Each local density pair is then $D(x\alpha)\tau_{n_c} D(x\alpha)^\dagger$, the same for both encodings. Signal displacement occupation is $2\alpha^2$ and noise occupation is $2n_c$ in each case. We set $\alpha = 1/2, n_c = 2$, so these budgets are $1/2$ and 4 photons. They are exact dimensionless model choices, not typical device specifications.

Theorem 3 (A certified bosonic reversal)**.** *For arbitrary local binary quantum measurements and a classical seed independent of both label and noise,*

$$C_{\text{align}} \geq L = \frac{1}{2} + \frac{7}{8}\frac{199}{1000} - \frac{25}{157} > 0.514889, \tag{15}$$

$$C_{\text{bal}} \leq U = \frac{1}{2} + \frac{100}{157}\frac{1/128}{1 - 128^{-8}} < 0.504977. \tag{16}$$

Nevertheless $P_{\text{align}} < 0.7$ *and* $P_{\text{bal}} > 0.9$*. These are analytical inequalities with exact rational certificates; no Fock cutoff or claim of an optimized numerical quantum measurement is used.*

4.1 A classical information relaxation bounds every quantum decoder

Grant each observer the exact coherent amplitude that prepared its local mode. It can simulate any local quantum effect's Born probability, so this is an information relaxation. The common imaginary amplitude is independent of x and the real noise and supplies only an extra shared seed. The balanced real-amplitude game is

$$w_1 = Z + x\alpha, \qquad w_2 = Z - x\alpha, \qquad Z \sim N(0, \sigma^2), \quad \sigma^2 = n_c/2. \tag{17}$$

Let deterministic functions f, g report $+1$ when equal to one. Allowing stochastic functions or shared mixtures cannot improve the best equal-prior average. The minimax score is bounded by that average.

Write $Z = r + \alpha + 2\alpha n$, $r \in [0, 2\alpha)$, and $p_n(r) = \varphi_\sigma(r + \alpha + 2\alpha n)$. For each residue, the positive-label event requires $f_{n+1} = g_n = 1$, while the negative-label event requires $f_n = g_{n+1} = 0$. Their prior-weighted contributions are $p_n/2$ each. The constraints split into two independent bi-infinite paths; adjacent rewarded edges conflict, and every nonadjacent set is realizable. Both paths have the weight sequence p_n. The factor two cancels the prior one half, giving the exact classical relaxed value

$$C_{\text{lat,bal}} = \int_0^{2\alpha} \text{MWIS}(p_n(r))\, dr, \tag{18}$$

where MWIS is the maximum sum of weights at nonadjacent integer indices. The constructive measurable parity optimizer below also justifies attaining the pointwise optimum.

For a positive summable strictly unimodal Gaussian sequence, allowing a two-site mode tie, an optimal independent set is one entire parity. Indeed, compactness and summability ensure an optimum; positive weights prevent termination at either end, and any gap at least four allows inserting a weight. If selected indices $i, i+3$ form a gap three, shift the selected left half one step right when the mode is at or right of $i+1$, or the right half one step left otherwise. Unimodality preserves or increases every moved weight, the boundary gap becomes two, and strict Gaussian

tails give a positive gain. Thus all gaps are two. Consequently MWIS is the maximum of the even and odd weight sums. To realize it measurably, let $e(r)$ choose the larger parity sum, breaking ties in favor of even. Set $f_n = 1$ on indices of parity opposite to $e(r)$ and $g_n = 1$ on indices of parity $e(r)$. These functions realize precisely the selected rewarded edges. The parity sums are continuous in r, so this choice is measurable.

Put $H(t) = \sum_j \varphi_\sigma(t + 4\alpha j)$ and $c = \sigma^2\pi^2/(8\alpha^2)$. Gaussian Fourier summation gives

$$H(t) = \frac{1}{4\alpha}\left[1 + 2\sum_{j\geq 1} e^{-cj^2}\cos\frac{j\pi t}{2\alpha}\right].$$

Using the two parity sums in (18) and integrating their absolute difference,

$$C_{\mathrm{bal}} \leq C_{\mathrm{lat,bal}} \leq \frac{1}{2} + \frac{2}{\pi}\sum_{j\geq 1\ \mathrm{odd}} e^{-cj^2} \leq \frac{1}{2} + \frac{2}{\pi}\frac{e^{-c}}{1-e^{-8c}}. \tag{19}$$

The integrated absolute value of each odd sine harmonic is $4\alpha/\pi$; no factor $1/j$ belongs in this upper bound. At our parameters $c = \pi^2/2$. The elementary bounds $157/50 < \pi < 22/7$ and the exact finite sum

$$\sum_{j=0}^{12} \frac{(24649/5000)^j}{j!} > 128$$

imply $e^c > 128$, proving $C_{\mathrm{bal}} \leq U$ in Theorem 3.

4.2 An elementary achievable lower and the pooled contrast

Aligned homodyne readouts in amplitude units are $Z + x\alpha + \nu_i$, with independent vacuum fluctuations $\nu_i \sim N(0, 1/4)$ in ideal homodyne outcomes. Zero-threshold decisions balance the two labels and give

$$C_{\mathrm{align}} \geq \mathbb{E}\Phi\left(\frac{\alpha + \min(\nu_1, \nu_2)}{\sigma}\right) \geq \Phi(\alpha/\sigma) - \frac{1}{2\pi\sigma}.$$

The second inequality uses the $1/\sqrt{2\pi}$ Lipschitz constant of Φ and $\mathbb{E}|\min(\nu_1, \nu_2)| = 1/\sqrt{2\pi}$. The latter follows by joint sign symmetry and $|\min| + |\max| = |\nu_1| + |\nu_2|$. With $\alpha/\sigma = 1/2$, monotonicity of the normal density gives $\Phi(1/2) \geq 1/2 + \varphi(1/2)/2$. The bounds $e^{-1/8} \geq 7/8$, $1/\sqrt{2\pi} > 199/500$ and $1/(2\pi) < 25/157$ prove $C_{\mathrm{align}} \geq L$. Exact rational subtraction yields $L - U > 0.0099132$.

A balanced two-mode transformation sends all common displacement into a bright thermal mode and all signal into a dark coherent mode. The exact pooled value is

$$P_{\mathrm{bal}} = \frac{1 + \sqrt{1 - e^{-8\alpha^2}}}{2} = \frac{1 + \sqrt{1 - e^{-2}}}{2} > 0.9. \tag{20}$$

For aligned signals, giving the pooled observer the latent amplitude is another information relaxation, so $P_{\mathrm{align}} \leq \Phi(\alpha/\sigma) = \Phi(1/2) < 0.7$. Thus the pooled ordering is strictly reversed. The mode decomposition is used only with pooled access, not granted to separated observers.

For context, direct one-dimensional quadrature evaluates the aligned homodyne probability as 0.580302 and the balanced exact-amplitude relaxation as 0.5045785. Neither is asserted to be the exact quantum consensus optimum. The rational proof, not these approximations, establishes the gap. Revealing Z through a seed independent only of x would invalidate the bound; the stated independence from preparation noise is essential.

5 Verification, interpretation and limits

The repository retains complete matrix constructors, attained upper bounds, exact certificate fractions, inputs and reproduction commands. Five qubit test groups check positivity, local marginal identity, source symmetry, closed radical values, global-bound attainment over a noise grid, arbitrary complex effects, and full-matrix probability expansions. An independent implementation imports none of that code and checks 101 noise points per encoding, including the one-way protocol; its largest matrix residual is below 8.4×10^{-16}.

A separate general-angle check covers 775 parameter pairs (1550 encoding cases), plus 100 arbitrary admissible frame pairs, and verifies the closed formulas, attained norm bound and one-way protocol. The largest evaluated upper-minus-attained discrepancy is below 2.3×10^{-16}. This is corroboration of the analytical open-region result, not a uniform numerical lower bound on its vanishing endpoint gaps.

The Gaussian certificate script uses standard-library integer and rational arithmetic to verify $L > U$. It additionally constructs a cellwise rational supremum bound of 0.508199 directly from finite-path dynamic programming and controlled omitted tails, without using the stronger parity or Fourier lemmas. That independent upper is still below L. Exhaustive small original decision graphs, independent-set calculations, and sampled stochastic responses test the path reduction. Numerical lattice quadrature, Gaussian integration and finite simulations are corroborating model checks, not laboratory repetitions or proof of global quantum optimization. Exact inequalities, floating-point residuals and heuristic searches are identified separately.

The physical message is an access-dependent encoder choice. Correlated preparation noise changes the relationship between independently correct reports and one aggregate decision even when every local density operator is fixed. Pooled discrimination is a valid criterion for its own task; it simply need not choose the encoder for a separated all-correct task. Conditional product assumptions would erase this mechanism and cannot be inferred from local density matrices.

This is not a demonstration of objective-state emergence, spectrum broadcast structure, a quantum advantage over all classical models, or an entanglement advantage. Generic task-ordering reversal is classical. The qubit classification covers pure product inputs and fixed unitary frames, not general CPTP processing or mixed/entangled encoders. The Gaussian proof gives bounds for two encodings rather than a complete encoder frontier. Shared entangled ancillas, communication in the consensus task, noise side information and postselection change the resource class. No hardware experiment or calibrated trapped-ion prediction is reported.

Open questions concern the completeness of broader encoder classes and the tightness of the Gaussian information relaxation.

A A classical comparator delimiting the conceptual claim

For a fair source bit, let both observers' conditional single-bit correctness be $q = 3/4$. Consider

$$P_0(a) = \begin{pmatrix} a & q-a \\ q-a & 1-2q+a \end{pmatrix}, \qquad P_1(b) = \begin{pmatrix} 1-2q+b & q-b \\ q-b & b \end{pmatrix}.$$

The choices $(a, b) = (3/4, 3/4)$ and $(1/2, 3/4)$ have exactly identical local conditional marginals. Exhausting the 16 deterministic local decoder pairs gives optimized Bayesian all-correct values $3/4$ and $5/8$, while the pooled values are $3/4$ and $7/8$. Private stochastic and shared mixtures cannot improve a linear average beyond a deterministic maximum. These values are an immediate application of the classical LSSD solution in Majenz et al. [8, Lemma 1], not a new abstract

phenomenon. The comparator as written is Bayesian and not label-symmetric; it is not used as a minimax proof. The matched-channel quantum construction provides the physically constrained, label-symmetric and completely optimized example studied in the main text.

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